The G-Nexus Utsira Project
Design of a 100% Renewable Off-Grid Energy System

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Contents

1 Abstract

This report presents the design and evaluation of a fully renewable, off-grid energy system for Utsira, Norway. The system is required to operate as a self-sufficient microgrid supplying residential demand, industrial activity, electric vehicle charging, water infrastructure, and a continuous 1 MW data center load.

Meteorological conditions are represented using datasets from the Frost API (MET Norway), Open-Meteo, and PVGIS, ensuring realistic modelling of wind and solar resources [6, 7, 8]. The proposed system consists of 5 MW wind capacity, 1 MW solar photovoltaic generation, and a 130 MWh battery energy storage system.

Hourly simulations show that the system meets total demand without any unmet load under normal operating conditions. A 48-hour Dunkelflaute scenario is also evaluated, confirming that the system maintains continuous supply under extreme low-generation conditions.

The results show that a fully renewable system is technically feasible, but requires significant oversizing of generation and storage. This leads to increased capital costs and energy curtailment, highlighting the trade-off between reliability and economic efficiency.

2 Introduction

The transition towards renewable energy systems introduces significant engineering challenges, particularly for isolated microgrids without access to external grid support. In such systems, electricity generation, storage, and demand must be balanced locally at all times, requiring robust design to ensure reliability under variable renewable conditions.

This project investigates the design of a fully renewable, off-grid energy system for Utsira, a small island municipality located off the western coast of Norway. According to Statistics Norway (SSB), Utsira has approximately 219 inhabitants and no defined urban settlement, indicating a dispersed population structure [1]. These characteristics directly influence both energy demand and infrastructure design.

Utsira has been identified as a relevant location for renewable energy development, particularly in wind power, aquaculture, and smart-grid testing [10]. The island has previously hosted wind energy projects and continues to be used as a test area for flexible energy systems, supporting the relevance of a renewable-based microgrid approach.

The system considered in this project must operate as a fully islanded microgrid following disconnection from the mainland grid. It is required to supply a continuous 1 MW data center load, residential electricity demand, industrial activity related to aquaculture and processing, electric vehicle charging infrastructure, and water and wastewater systems.

The inclusion of the data center introduces a dominant and continuous base load. Data centers are characterized by near-constant power demand and high reliability requirements, significantly increasing system design constraints [19].

A critical requirement defined in the assignment is the ability to withstand a 48-hour *Dunkelflaute* event, defined as a period with simultaneously low wind and solar generation. Such events represent worst-case scenarios for renewable energy systems and must be explicitly considered in system design.

The objective of this project is therefore to design and evaluate a fully renewable, off-grid energy system capable of supplying all electricity demand on Utsira with high reliability, including during extreme low-generation events. To achieve this, the study combines local socio-economic data, meteorological datasets, and engineering assumptions to model load demand, renewable generation, and energy storage requirements.

3 Local Conditions and Assumptions

3.1 Geographical and Socio-Economic Context

Utsira is a small island municipality located in the North Sea, west of mainland Norway. Statistics Norway (SSB) reports that the municipality has around 219 inhabitants and no defined urban settlement, indicating a highly dispersed population [1]. This has direct implications for infrastructure design, including limited potential for centralized heating systems and short transportation distances.

The island is strongly exposed to maritime weather conditions, making it particularly suitable for wind energy generation. Utsira municipality identifies wind power as a key local resource and has previously hosted operational wind turbines with significant annual production [9].

In addition to renewable energy, Utsira is developing within aquaculture and maritime industries [11]. These activities introduce additional electricity demand, particularly for processing and cooling systems. The municipality is also used as a test area for smart-grid and flexibility solutions, indicating relevance for energy storage and demand-side management technologies [10].

3.2 Energy Demand Structure

The energy system must supply multiple demand sectors with different temporal characteristics. The data center represents a continuous base load, while residential demand varies seasonally due to electric heating. Industrial demand is assumed to be relatively stable, reflecting aquaculture and processing activities. Electric vehicle charging introduces time-dependent demand, typically peaking in the evening, while water and wastewater systems contribute a smaller but continuous load.

This diversity requires a system capable of handling both short-term fluctuations and long-term seasonal variation.

3.3 Key Model Parameters

All model inputs are defined in Table 1. Each parameter is classified as measured data, official statistics, or engineering assumptions. This distinction is critical to ensure transparency and traceability, as required by the assignment.

Table 1: Key model parameters

Parameter	Value	Unit	Source	Type
Population	219	residents	[1]	Official
Households	88–100	hh	[2]	Official
Residential demand	15–18	MWh/hh·yr	[3]	Official
Data center load	1	MW	[20]	Measured
EV chargers	200	units	[20]	Measured
EV simultaneity	≤40%	–	[16,17]	Literature
Industrial SEC	85–132	kWh/ton	[12,13]	Literature
Industrial load	20–50	kW	–	Assumption
Water energy	0.3–0.8	kWh/m^3	[14,15]	Literature
Wind resource	>5	m/s	[6]	Measured
Solar yield	∼900	kWh/kWp·yr	[8]	Measured
Battery capacity	130	MWh	–	Assumption
Battery efficiency	∼92%	–	[18]	Literature

3.4 Critical Assessment of Assumptions

Most key parameters are based on measured data or official statistics, ensuring a high level of credibility. However, several inputs rely on engineering assumptions and must be treated with caution.

The industrial load is not based on measured data from Utsira, but derived from literature values and an assumed processing scale. Similarly, water and wastewater energy use is based on generic literature ranges rather than municipality-specific data. Battery sizing is also an engineering decision derived from system requirements rather than external datasets.

These assumptions introduce uncertainty into the model and should be interpreted as first-order estimates. Their impact on system performance must therefore be considered when evaluating the results.

4 Meteorological Data and Renewable Resource Assessment

4.1 Data Sources and Methodology

Accurate meteorological data is essential for realistic modelling of renewable energy systems. In this study, weather data is obtained from the Frost API provided by the Norwegian Meteorological Institute, which provides historical observations of wind speed, temperature, and atmospheric conditions [6].

Solar-related data is obtained from Open-Meteo and validated against PVGIS datasets from the European Commission, which provide location-specific solar radiation and photovoltaic (PV) yield estimates [7, 8].

The use of measured meteorological data ensures that both seasonal variation and short-term fluctuations are captured. This is critical for evaluating system performance under realistic operating conditions, including extreme events relevant for system reliability.

4.2 Wind Resource Assessment

Utsira is located in a coastal region with strong exposure to North Sea wind conditions. Observations from the Frost dataset indicate that wind speeds frequently exceed 5 m/s, which is generally considered the lower threshold for viable wind power production.

Local evidence further supports this. Historical wind turbines on Utsira have produced approximately 3.5 GWh annually with around 3500 operating hours per year [9]. This corresponds to a relatively high capacity factor and confirms that wind is a reliable energy resource in the region.

The power available in wind is given by:

$$P = \frac{1}{2}\rho A v^3 C_p \tag{1}$$

The cubic dependence on wind speed implies that even moderate increases in wind speed lead to significant increases in power output. Based on both measured data and local evidence, a representative average wind speed of approximately 5–6 m/s is assumed for modelling purposes.

Typical capacity factors for onshore wind in comparable Norwegian coastal regions range between 30% and 45% [9]. This makes wind the most reliable renewable resource for Utsira, particularly during winter months when electricity demand is highest.

4.3 Solar Resource Assessment

Solar irradiance is obtained from Open-Meteo and validated using PVGIS data. PVGIS indicates an annual photovoltaic yield of approximately 900 kWh per kWp installed capacity for Norway [8].

The solar resource exhibits strong seasonal variation, with very limited production during winter months due to low solar angles and short daylight hours. In contrast, summer months provide significantly higher irradiance and longer periods of generation.

This seasonal mismatch is critical, as peak electricity demand occurs during winter when solar production is lowest.

4.4 Complementarity of Wind and Solar

The combined behaviour of wind and solar resources shows a clear asymmetry. Wind energy is available throughout the year and tends to be stronger during winter, while solar energy is highly seasonal and contributes mainly during summer and daytime periods.

This complementary behaviour supports a hybrid generation system, where wind provides the primary energy supply and solar reduces system stress during favorable conditions.

4.5 Implications for System Design

The meteorological analysis has direct implications for system configuration. Wind power must serve as the backbone of the system due to its availability and alignment with peak demand. Solar PV should be implemented as a secondary source to provide additional generation during summer and reduce reliance on storage.

Despite this complementarity, both resources are variable and non-dispatchable. This introduces the need for energy storage to ensure continuous supply, particularly during periods of low generation.

5 Load Modelling

The total electricity demand on Utsira is composed of several sectors with different temporal characteristics. The overall system load is defined as:

$$P_{\mathrm{load}}(t) = P_{\mathrm{DC}}(t) + P_{\mathrm{res}}(t) + P_{\mathrm{EV}}(t) + P_{\mathrm{ind}}(t) + P_{\mathrm{water}}(t) \tag{2}$$

Each component is modelled separately based on a combination of measured data, official statistics, and engineering assumptions.

5.1 Data Center Load

The data center represents the dominant load in the system and is treated as critical infrastructure. The load profile is obtained directly from the provided G-Nexus dataset, which serves as the primary measured data source.

The average load is approximately 1 MW, corresponding to an annual energy consumption of:

$$E_{\mathrm{DC}} = 8760 \text{ MWh/year} \tag{3}$$

Data centers are characterized by continuous operation and high reliability requirements, making them the defining constraint for system design [19].

5.2 Residential Load

Residential demand is based on population and household data from Statistics Norway. Utsiras population is spread at a estimated 88–100 households [1, 2].

Average electricity consumption in Norwegian households is approximately 15–18 MWh per year, reflecting the widespread use of electric heating [3]. This value is used as a baseline for modelling.

The temporal variation in residential demand is strongly influenced by temperature. Electricity consumption increases significantly during winter due to heating requirements, with peak demand typically occurring during morning and evening periods [5]. Heating demand is therefore modelled as a function of ambient temperature, introducing seasonal variability into the load profile.

5.3 Electric Vehicle Load

The system includes 200 EV charging points as specified in the assignment. Each charger has a rated power of 7.4 kW, resulting in a theoretical maximum load of 1.48 MW.

However, simultaneous usage of all chargers is unrealistic. Studies of Norwegian charging behaviour indicate that simultaneity factors are typically below 40% [17, 16]. The effective peak load is therefore modelled as a fraction of the installed capacity.

Charging demand is time-dependent, with peak usage typically occurring in the evening when users return home. Due to the small geographical size of Utsira and limited road infrastructure, daily energy consumption per vehicle is assumed to be relatively low, while peak power demand may still remain significant.

5.4 Industrial Load

Industrial demand is associated with aquaculture and fish processing activities. Energy consumption in such processes is typically reported in the range of 85–132 kWh per ton of processed fish [12, 13].

Due to the lack of specific production data for Utsira, industrial demand is modelled as a constant load based on an assumed processing scale. This represents a proxy estimate and introduces uncertainty into the model.

5.5 Water and Wastewater Systems

Water supply and wastewater treatment contribute a small but continuous load. Typical energy intensities are reported as approximately 0.3 kWh/m^3 for water supply and 0.5–0.8 kWh/m^3 for wastewater treatment [14, 15].

In the absence of municipality-specific data, these values are used as representative estimates scaled according to population.

5.6 Load Composition and System Implications

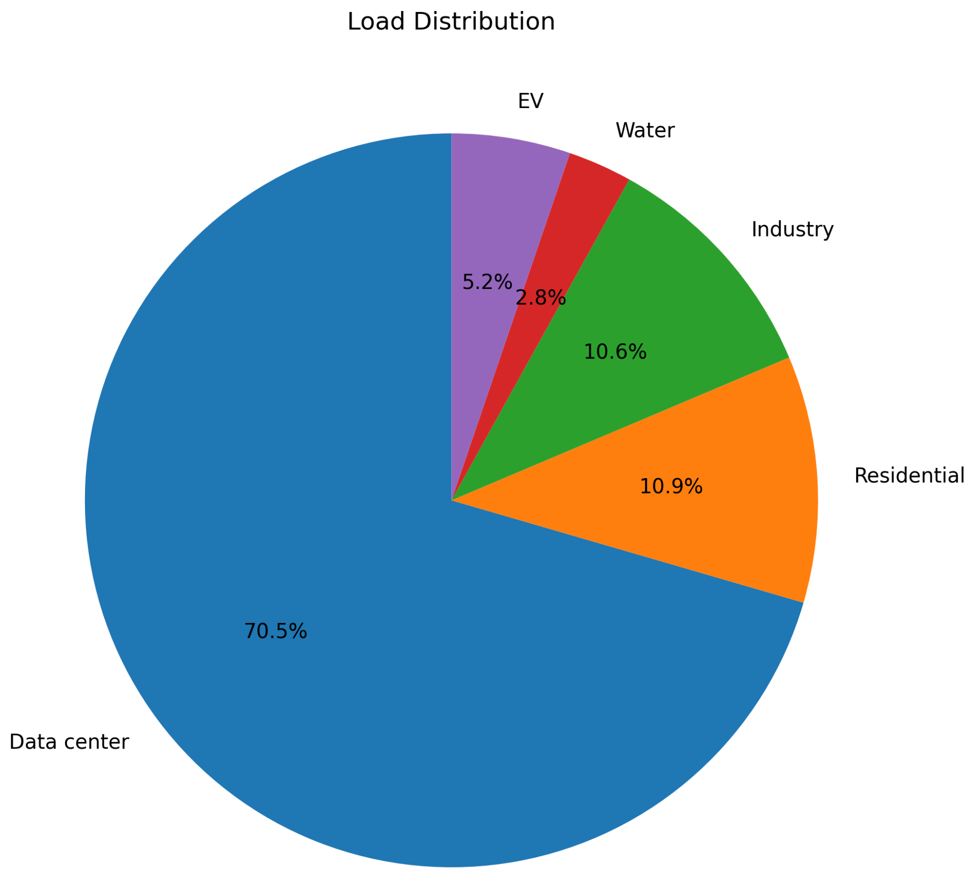

Figure 1: Load distribution by sector

The resulting load composition is dominated by the data center, which accounts for the majority of total energy demand. Residential demand introduces seasonal variation, while EV charging and industrial loads contribute to peak demand.

This structure results in a system that is strongly base-load dominated. As a consequence, reliability requirements are driven primarily by the continuous data center demand, while storage requirements are determined by the need to sustain this load during low-generation periods.

5.7 Critical Assessment of Load Modelling

The confidence level of the load components varies significantly. The data center load is based on measured data and therefore has high reliability. Residential demand is supported by national statistics and has moderate confidence. In contrast, industrial and EV loads rely on proxy assumptions and introduce greater uncertainty.

These uncertainties must be considered when interpreting the simulation results, as they directly affect both system sizing and performance evaluation.

6 Renewable Generation System Design

The generation system is designed based on local resource conditions and the requirement for high reliability in an isolated microgrid. Due to the absence of grid support, generation must not only meet annual demand but also provide sufficient surplus to charge storage and maintain supply during low-generation periods.

The system consists of wind power as the primary energy source and solar photovoltaic (PV) as a secondary, complementary source.

6.1 Wind Power as Primary Energy Source

Wind energy is the dominant renewable resource on Utsira due to its exposure to North Sea weather systems. Historical data and local evidence confirm strong and consistent wind conditions. The municipality reports that earlier wind turbines achieved approximately 3.5 GWh annual production with around 3500 operating hours per year, indicating a high capacity factor relative to typical onshore installations [9].

Meteorological data from the Frost API further supports this, showing frequent wind speeds above 5–6 m/s and stronger conditions during autumn and winter [6]. This seasonal alignment is critical, as electricity demand is highest during winter months [5].

Wind power is therefore selected as the backbone of the system.

6.2 Wind Capacity Selection

The installed wind capacity is defined as:

$$P_{\text{wind}} = 5 \text{ MW} \tag{4}$$

This value is derived from the requirement to cover both average demand and storage needs. Assuming a capacity factor of approximately 40%, annual wind production becomes:

$$E_{\text{wind}} = 5 \cdot 8760 \cdot 0.40 = 17{,}520 \text{ MWh/year} \tag{5}$$

This exceeds the annual data center demand of 8760 MWh and contributes to covering residential and additional loads.

The capacity is intentionally oversized to ensure sufficient generation during low-wind periods and to allow charging of the battery system. This is a standard design strategy in isolated renewable systems where reliability is prioritized over efficiency [18, 5].

6.3 Solar Photovoltaic as Complementary Source

Solar PV is included to supplement wind generation. Data from PVGIS indicates an annual yield of approximately 900 kWh per kWp in Norway, with strong seasonal variation [8]. Production is concentrated in summer months and is negligible during winter.

This seasonal mismatch means that solar cannot serve as a primary energy source. Instead, it contributes by reducing demand on the battery during periods of high irradiance.

6.4 Solar Capacity Selection

The installed solar capacity is defined as:

$$P_{\text{solar}} = 1 \text{ MW} \tag{6}$$

This value reflects a balance between contribution and practicality. Given the limited land area of Utsira, solar installations are assumed to be distributed and primarily rooftop-based. Solar PV also has lower installation costs compared to wind power, making it a cost-effective supplementary technology [18].

Higher solar penetration was considered but rejected due to its limited contribution during winter and the resulting increase in required storage capacity.

6.5 Combined Generation Behaviour

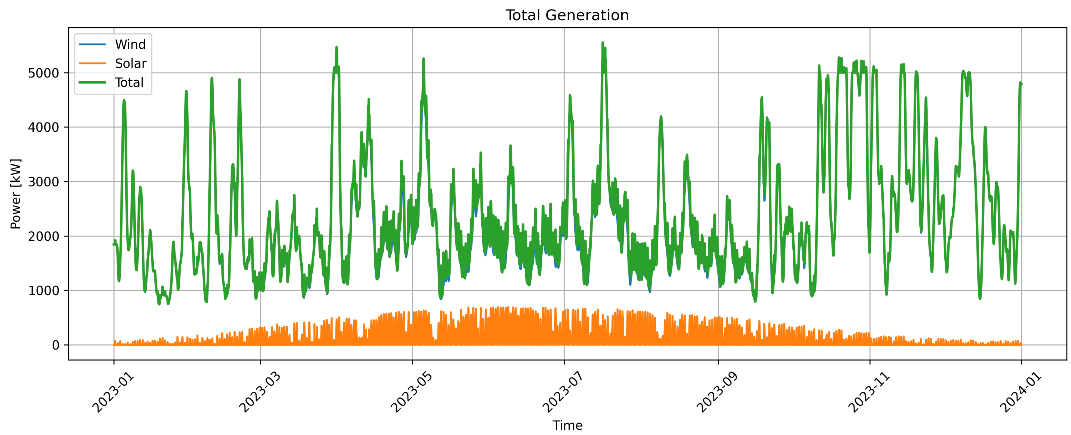

Figure 2: Total renewable generation (wind and solar)

The combined generation profile is characterized by wind dominance throughout the year and solar contribution during summer. The system frequently produces surplus energy due to the oversizing of generation capacity.

This surplus is necessary for charging the battery system but also leads to curtailment when storage capacity is exceeded.

6.6 Engineering Trade-Offs

The selected generation mix results in a system that prioritizes reliability. Oversizing ensures sufficient energy availability during unfavorable conditions but introduces inefficiencies in the form of curtailed energy.

A higher share of solar would reduce summer curtailment but significantly weaken winter performance. Conversely, reducing wind capacity would increase the risk of unmet load during low-wind periods.

The chosen configuration therefore represents a compromise between reliability, resource availability, and system constraints.

6.7 Summary of Generation Design

The final generation system is defined by:

- Wind power (5 MW) as the primary energy source
- Solar PV (1 MW) as a complementary source

This configuration is justified by local meteorological conditions, seasonal demand patterns, and the requirement for high system reliability in an isolated environment.

7 Battery Energy Storage System (BESS)

7.1 Role of Energy Storage

In an isolated renewable energy system, energy storage is required to balance the mismatch between generation and demand. Wind and solar power are non-dispatchable, meaning that electricity cannot be produced on demand. The battery therefore ensures continuous supply by storing surplus energy and delivering it during periods of deficit.

For Utsira, storage is the only mechanism available to maintain system reliability during low-generation events.

7.2 System Configuration

The system includes a lithium-ion battery with the following parameters:

- Energy capacity: 130 MWh
- Power rating: 6 MW
- Round-trip efficiency: 92% [18]
- Initial state of charge: 60–80%

The battery state of charge evolves as:

$$E(t+1) = E(t) + \eta P_{\text{charge}}(t) - \frac{P_{\text{discharge}}(t)}{\eta} \tag{7}$$

with:

$$0 \leq E(t) \leq E_{\max} \tag{8}$$

7.3 Energy Capacity Sizing

The battery is primarily sized to ensure system survival during a 48-hour Dunkelflaute event. Under this condition, renewable generation is assumed to be negligible and the system must rely entirely on stored energy.

The required energy is estimated as:

$$E_{\text{req}} = P_{\text{critical}} \cdot t \tag{9}$$

Assuming a critical load of approximately 1.2 MW:

$$E_{\text{req}} = 1.2 \cdot 48 = 57.6 \text{ MWh} \tag{10}$$

Accounting for efficiency losses:

$$E_{\text{req}} = \frac{57.6}{0.92} \approx 62.6 \text{ MWh} \tag{11}$$

This represents the theoretical minimum. In practice, additional capacity is required to account for uncertainty in load modelling, variability in real weather conditions, and battery degradation over time.

A safety factor of approximately 2 is therefore applied, resulting in a design capacity of:

$$E_{\text{battery}} = 130 \text{ MWh} \tag{12}$$

This ensures that the system can operate without requiring a fully charged battery and maintains a sufficient reliability margin.

7.4 Power Rating Selection

The battery power rating determines how quickly energy can be delivered or absorbed. The system must be able to respond to short-term fluctuations in wind generation and cover peak demand.

The peak system load is approximately 1–2 MW. A power rating of 6 MW is therefore selected to ensure:

- Sufficient discharge capability during sudden drops in generation
- Fast charging during periods of surplus
- No power limitation during peak demand

This separation between energy capacity (MWh) and power rating (MW) is essential for stable system operation.

7.5 System Operation

At each timestep, the system satisfies the power balance:

$$P_{\text{generation}}(t) + P_{\text{battery}}(t) = P_{\text{load}}(t) \tag{13}$$

The battery operates according to a rule-based dispatch strategy. When generation exceeds demand, surplus energy is stored. When demand exceeds generation, the battery discharges to compensate. If the battery reaches full capacity, excess energy is curtailed. If the battery is depleted, unmet load occurs.

7.6 Simulation Results

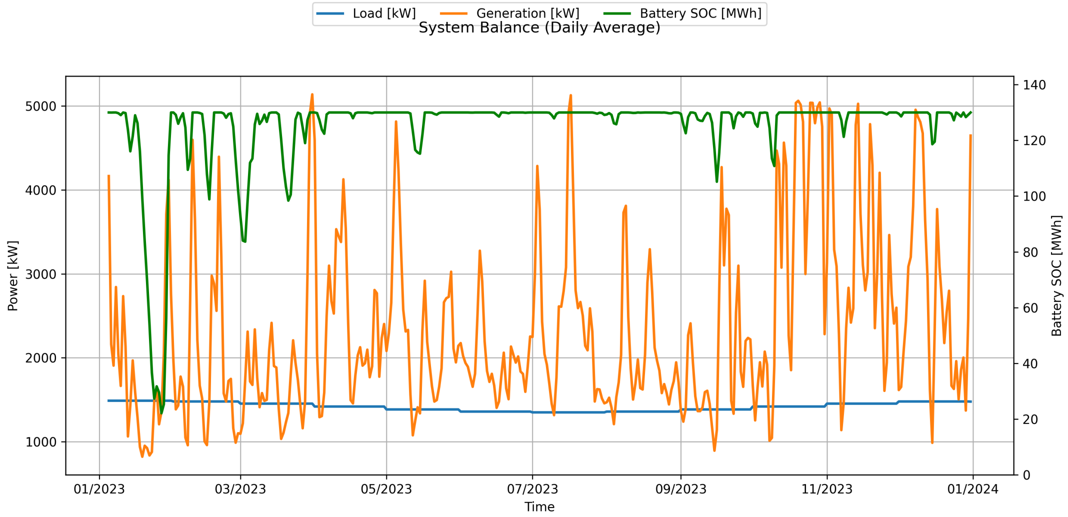

Figure 3: System balance showing generation, load, and battery state of charge

The results show that the battery actively balances the system, charging during high wind periods and discharging during deficits. Seasonal trends are visible, with increased discharge during winter due to higher demand and reduced solar contribution.

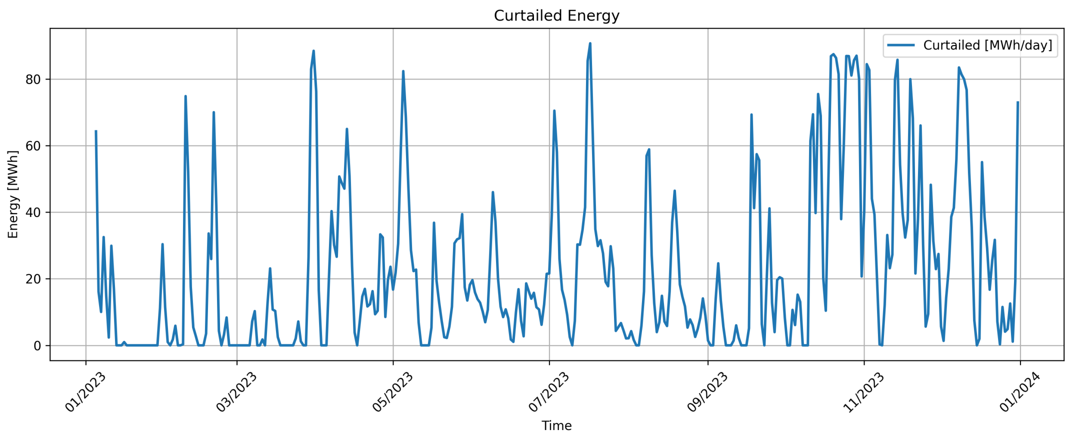

Figure 4: Curtailed renewable energy

Curtailment refers to the reduction or rejection of available renewable energy when generation exceeds demand and storage capacity. Frequent curtailment occurs due to the oversizing of generation capacity. While this reduces system efficiency, it is a necessary trade-off to ensure reliability in an isolated system.

7.7 Unmet Load Verification

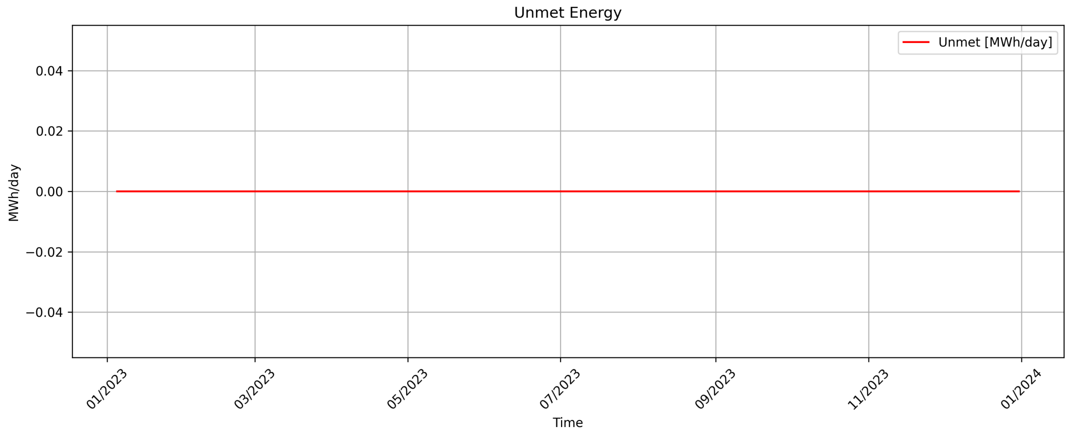

Figure 5: Unmet load over time

The simulation shows negligible unmet load under normal operating conditions, confirming that the selected battery capacity is sufficient to maintain continuous supply.

7.8 Engineering Assessment

The battery system is the key enabling component of the energy system. Its energy capacity determines the ability to survive long-duration deficits, while its power rating ensures stability during short-term fluctuations.

The selected configuration prioritizes reliability through conservative sizing. However, this introduces high capital cost and frequent cycling, which may impact long-term performance due to degradation.

Overall, the design represents a trade-off between robustness and economic efficiency, consistent with the requirements of an isolated microgrid.

8 Dunkelflaute Analysis and System Reliability

8.1 Definition and Relevance

A Dunkelflaute event refers to a period of simultaneously low wind speeds and minimal solar irradiance, typically occurring during winter in Northern Europe [21]. These events represent the most critical operating condition for renewable energy systems.

For an isolated microgrid such as Utsira, where no external grid support is available, system reliability depends entirely on local generation and storage. The system must therefore maintain:

$$P_{\text{unmet}}(t) = 0 \quad \forall t \tag{14}$$

even under extreme conditions.

8.2 Scenario Definition

The system is evaluated under a worst-case design scenario defined by the assignment requirements:

- Duration: 48 hours
- Wind generation: negligible
- Solar generation: negligible
- Load: winter conditions

The battery is initialized at:

$$SOC_{\text{initial}} = 80\% \tag{15}$$

This reflects a realistic operating strategy where storage is charged ahead of expected low-generation periods.

8.3 Simulation Results

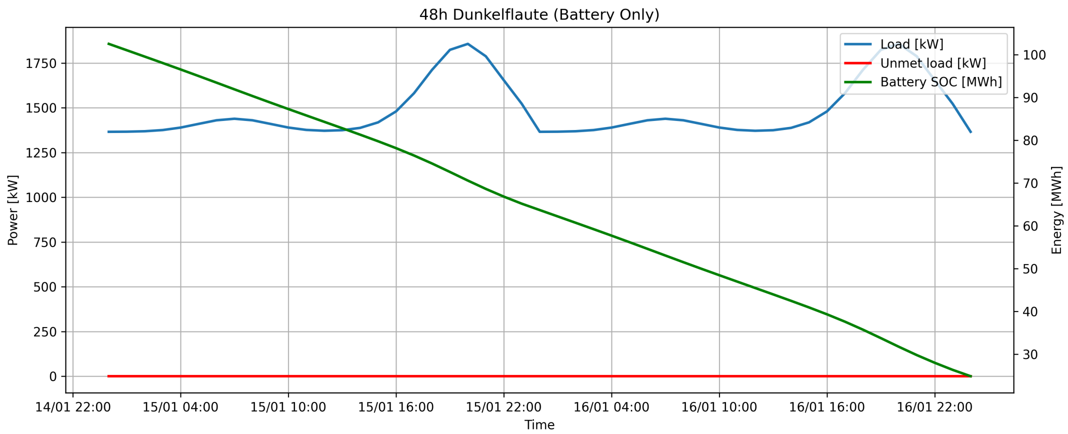

Figure 6: System performance during a 48-hour Dunkelflaute event

The results show that the system maintains continuous supply throughout the event, with no unmet load observed. The battery state of charge decreases approximately linearly, reflecting the dominance of a stable base load.

8.4 Engineering Interpretation

The results confirm that the battery energy capacity is sufficient to sustain system operation during the design-level extreme event. The system does not require a fully charged battery at the start of the event, indicating the presence of a safety margin.

This demonstrates that system reliability is primarily determined by energy capacity (MWh), while power rating (MW) is sufficient to meet instantaneous demand.

8.5 Limitations of the Scenario

The adopted scenario represents a conservative but simplified worst-case condition. In reality, Dunkelflaute events typically involve low but non-zero generation, and their duration can vary.

The analysis is therefore best interpreted as a design verification rather than a complete representation of all possible operating conditions.

8.6 Reliability Assessment

The system satisfies the critical requirement of continuous supply during a 48-hour Dunkelflaute event. This confirms that the selected combination of generation and storage provides sufficient robustness for isolated operation under extreme conditions.

9 Hydrogen Storage Evaluation

Hydrogen storage was evaluated as a potential alternative to battery storage for balancing renewable generation. A hydrogen-based system typically consists of an electrolyzer, storage tank, and fuel cell, enabling conversion between electricity and hydrogen.

The primary advantage of hydrogen is its suitability for long-duration and seasonal energy storage. However, this comes at the cost of significantly lower round-trip efficiency compared to batteries.

Typical round-trip efficiencies are:

$$\eta_{\text{hydrogen}} \approx 40\text{–}50\% \tag{16}$$

$$\eta_{\text{battery}} \approx 85\text{–}95\%[18] \tag{17}$$

This implies that more than half of the input energy is lost in a hydrogen system, whereas battery systems retain the majority of stored energy.

For the Utsira system, the required storage duration is on the order of 48 hours. At this timescale, efficiency losses dominate system performance. A hydrogen-based solution would therefore require significantly larger generation capacity to compensate for conversion losses, increasing both cost and system complexity.

In addition, hydrogen systems introduce further challenges related to infrastructure, safety, and operational complexity, particularly for a small-scale isolated system.

Hydrogen storage is therefore not considered suitable for this application. While it may be relevant for seasonal storage in larger energy systems, battery storage remains the most appropriate solution for short-duration reliability requirements.

10 District Heating and Waste Heat Utilization

Data centers convert nearly all consumed electrical energy into heat due to server operation and cooling processes. According to the International Energy Agency (IEA), this waste heat represents a significant potential for energy recovery [19].

For a data center with a continuous load of 1 MW, the theoretical heat output is approximately:

$$Q_{\text{heat}} \approx 1 \text{ MW} \tag{18}$$

In practice, only a fraction of this heat can be recovered due to system losses and temperature limitations. A typical recoverable share is assumed to be approximately 60%, resulting in:

$$Q_{\text{usable}} \approx 0.6 \text{ MW} \tag{19}$$

This recovered heat can be used to offset electrical heating demand in residential buildings and public infrastructure. Since heating represents a major component of electricity consumption in Norway, particularly during winter [5], waste heat utilization can reduce peak electrical load and improve overall system performance.

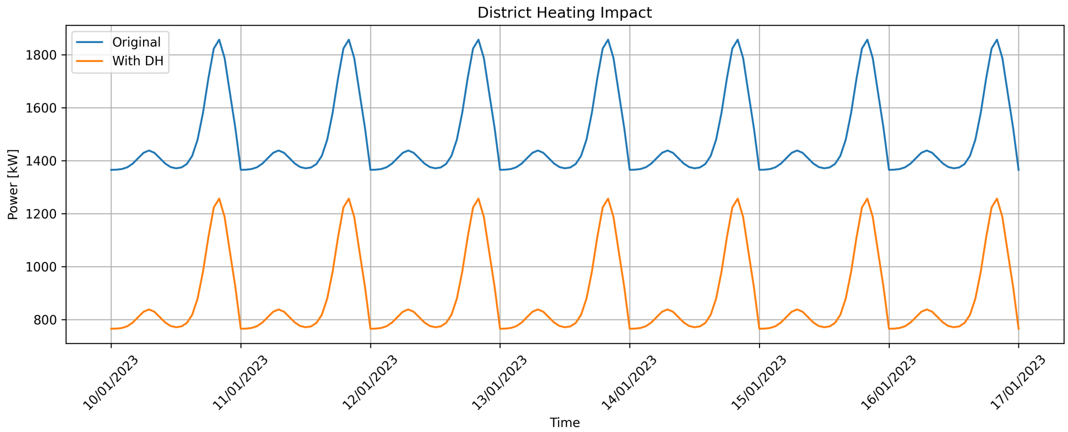

Figure 7: Reduction in electrical load due to waste heat utilization

The results indicate that heat recovery reduces winter peak demand and decreases the required discharge from the battery system. This contributes to improved system stability and reduced stress on storage.

However, the practical implementation of district heating on Utsira is constrained by local conditions. The municipality has a dispersed population structure and no dense urban settlement [1], which limits the feasibility of large-scale heat distribution networks.

In addition, data center waste heat is typically low-temperature (20–40°C), which may require heat pumps or low-temperature heating systems to be effectively utilized. This introduces additional complexity and energy consumption.

As a result, full-scale district heating is unlikely to be economically viable. A more realistic approach is localized heat reuse, where nearby buildings directly utilize recovered heat.

Overall, waste heat recovery represents a valuable opportunity to improve system efficiency and reduce electrical demand, but its practical impact on Utsira is limited by geographical and technical constraints.

11 Economic Analysis

11.1 Methodology and Limitations

The economic evaluation is based on international benchmark data due to the lack of project-specific cost information for Utsira. Cost estimates are derived primarily from IRENA reports on renewable energy technologies [18].

All values should therefore be interpreted as order-of-magnitude estimates rather than precise investment costs.

11.2 Capital Expenditure

The main system components are wind power, solar PV, and battery storage. Typical cost ranges are:

Table 2: Estimated CAPEX for system components

Component	Capacity	Unit Cost	Total Cost
Wind Power	5 MW	1300–1800 €/kW	6.5–9.0 M€
Solar PV	1 MW	800–1200 €/kW	0.8–1.2 M€
Battery Storage	130 MWh	200–400 €/kWh	26–52 M€
Total	–	–	33–62 M€

The battery system dominates total investment cost, accounting for the majority of CAPEX. This reflects the high cost of energy storage relative to generation technologies.

11.3 Operational Considerations

Operational costs for renewable systems are relatively low compared to capital costs. Wind and solar typically require 1–4% of CAPEX annually, while battery systems incur additional costs related to degradation and replacement [18].

Battery lifetime is typically 10–15 years, depending on cycling behaviour. The high utilization observed in the simulation suggests that degradation may be a significant cost factor over time.

11.4 Cost Drivers

The overall system cost is strongly influenced by three key factors: battery cost per kWh, required storage capacity, and wind resource quality.

A reduction in battery cost would significantly improve economic feasibility, while lower wind production would increase required storage capacity and therefore total cost. This highlights the sensitivity of the system to both technology development and local resource conditions.

11.5 Reliability–Cost Trade-Off

The system is intentionally designed to prioritize reliability, resulting in oversized generation capacity and storage. This ensures continuous supply but leads to higher capital cost and energy curtailment.

This represents a fundamental trade-off in isolated renewable systems. Reducing system size would improve economic efficiency but increase the risk of unmet load during extreme conditions.

For critical infrastructure such as a data center, the cost of downtime may exceed the additional cost of oversizing, justifying a reliability-focused design.

11.6 Utsira-Specific Considerations

Additional costs not included in the estimates may further increase total investment. These include transportation of equipment to an island location, installation challenges in coastal conditions, and maintenance logistics.

These factors introduce additional uncertainty and should be considered in a more detailed economic analysis.

11.7 Summary

The proposed system is technically feasible but capital-intensive, with battery storage representing the dominant cost component. Economic viability depends strongly on future cost reductions in storage technologies and the value placed on reliability and energy independence.

12 Discussion

12.1 Technical Feasibility

The results demonstrate that a fully renewable, off-grid energy system for Utsira is technically feasible under the defined assumptions. The system meets total demand throughout the year and maintains continuous supply during the 48-hour Dunkelflaute scenario.

Wind power is confirmed as the dominant energy source due to strong and seasonally aligned resource availability. Solar PV provides complementary support during summer, while the battery system enables continuous operation by compensating for variability in generation.

However, this feasibility is conditional on the assumptions used in the model and does not represent a guaranteed real-world outcome.

12.2 Impact of Oversizing

The system is intentionally oversized in both generation and storage capacity to ensure reliability. This leads to frequent energy surplus and curtailment, which reduces overall system efficiency.

This reflects a fundamental trade-off in isolated renewable systems: increasing reliability requires additional capacity, while reducing cost increases the risk of supply shortages. The selected design therefore represents a conservative solution rather than an optimized one.

12.3 Uncertainty and Assumptions

Several key aspects of the model introduce uncertainty. Residential demand is based on national averages rather than local measurements, and industrial and EV loads are derived from proxy assumptions. Meteorological modelling is limited to a single year of data, which may not capture extreme long-term variability.

The Dunkelflaute scenario is also simplified, assuming zero generation over a fixed duration. While this provides a clear design benchmark, it does not represent all possible real-world conditions.

Battery degradation and long-term performance are not explicitly modelled, which may affect system reliability and cost over time.

12.4 System Limitations

The study is limited by the availability of local data and the use of simplified modelling approaches. In particular, the lack of measured non–data center electricity consumption introduces uncertainty in total load estimation.

In addition, the economic analysis is based on general cost benchmarks and does not include detailed financial modelling or site-specific cost factors.

These limitations should be considered when interpreting the results.

12.5 Potential Improvements

The current system design prioritizes robustness but is not necessarily optimal. Improvements could include the use of multi-year meteorological datasets, more accurate local demand data, and optimization of generation and storage sizing.

Further development could also explore demand-side flexibility, such as controlled EV charging, to reduce peak load and storage requirements.

12.6 Overall Assessment

The system represents a technically sound solution for supplying an isolated community with renewable energy. It achieves high reliability through conservative design choices, but at the cost of efficiency and economic performance.

The results highlight that the main challenge in such systems is not resource availability, but the balance between reliability, cost, and system design complexity.

12.7 Project Execution and Group Contribution

The project was carried out through a division of tasks based on individual strengths and contributions.

Ikram contributed to the conceptual development of the project, including the justification and reasoning behind the selected system design. She also played a central role in preparing the presentation and in gathering and visualizing data used for communication of results.

Daniel was responsible for sourcing and verifying relevant literature and references. He contributed to the report by reviewing content, ensuring consistency, and supporting proofreading and quality control.

André was responsible for the technical implementation of the project. This included development of all simulation models, data processing, and integration of external datasets. In addition, André coordinated the overall structure of the project after the initial concept phase and was responsible for assembling and writing the main body of the report.

Throughout the project, Daniel and André worked together to address technical challenges that arose during the development process. This included discussions on key design decisions, such as battery sizing, system configuration, and performance trade-offs, contributing to more robust and well-justified solutions.

While the division of work allowed efficient progress, the workflow could have benefited from more continuous collaboration and earlier integration of tasks during the report development phase.

13 Conclusion

This study has demonstrated that a fully renewable, off-grid energy system for Utsira is technically feasible under the defined assumptions and modelling framework. The proposed system, consisting of 5 MW wind power, 1 MW solar photovoltaic generation, and 130 MWh battery storage, is capable of supplying total electricity demand throughout the year without significant unmet load.

Wind power is identified as the dominant energy source due to strong and seasonally aligned resource availability, particularly during winter when demand is highest. Solar PV contributes as a complementary resource, primarily reducing system stress during summer and daytime periods.

The battery system is the key enabling component, ensuring continuous operation by balancing variability in renewable generation. The system successfully satisfies the critical requirement of maintaining supply during a 48-hour Dunkelflaute event.

However, this level of reliability is achieved through significant oversizing of both generation and storage capacity. This results in high capital costs and energy curtailment, illustrating the fundamental trade-off between reliability, efficiency, and economic feasibility in isolated renewable energy systems.

Waste heat recovery from the data center provides additional benefits by reducing electrical heating demand, although its practical implementation is limited by local geographical conditions.

Overall, the results indicate that the primary challenge in designing isolated renewable systems is not resource availability, but achieving an appropriate balance between system reliability, cost, and design complexity.

14 AI Audit

Artificial Intelligence (AI) tools were used as a supporting resource during this project.

AI was primarily used for:

- Structuring the report and improving academic writing
- Identifying and organizing relevant sources
- Debugging code and assisting with API usage (Frost, Open-Meteo)
- Supporting interpretation of results and engineering reasoning

All modelling, simulations, and engineering decisions were developed and verified by the authors. External sources have been critically evaluated and used to justify assumptions where direct data was not available.

AI has not been used to generate results or replace engineering judgement, but rather as a tool to support analysis, improve clarity, and ensure consistency throughout the report.

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